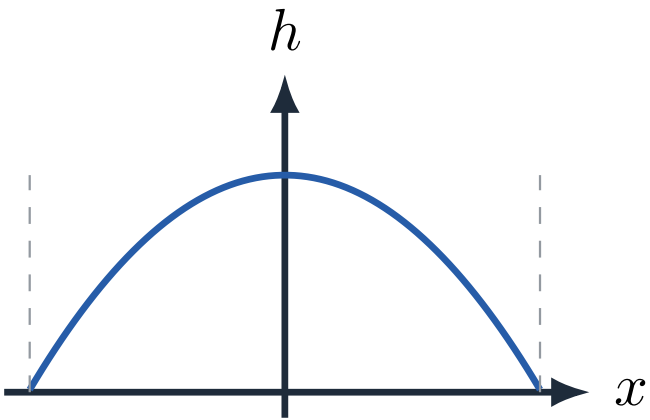
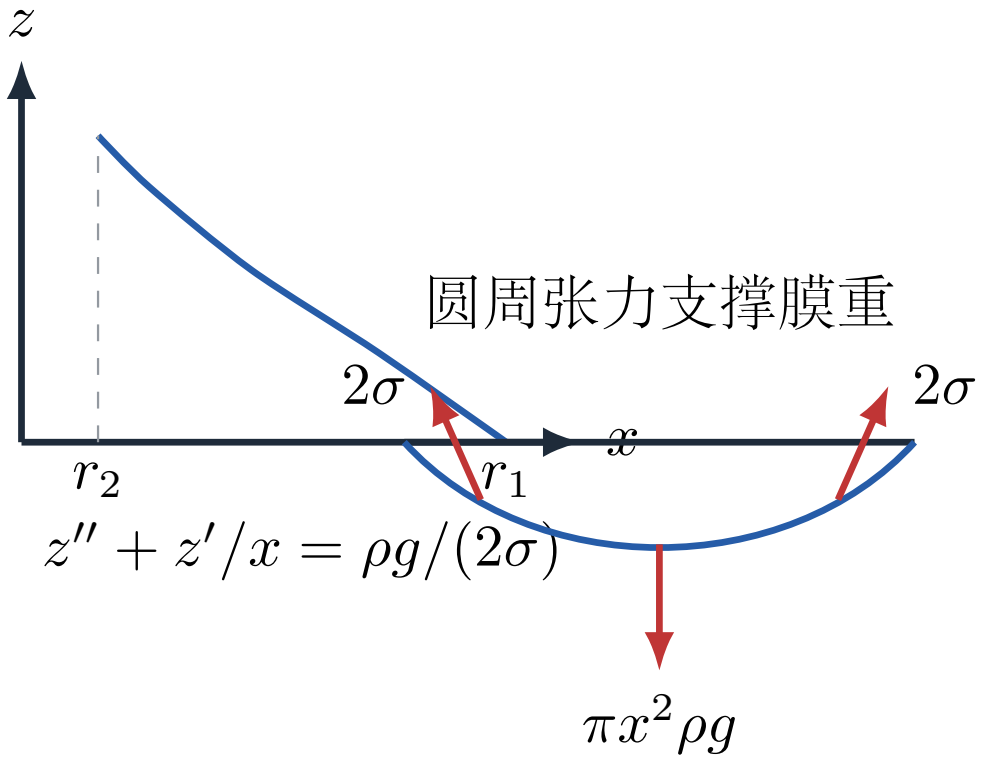
$$h(x) = \frac{\rho g}{8\sigma}(r^2 - x^2)$$